

```

#include <stdio.h>
#include <string.h>

#define MAX_RESERVATIONS 1000

// structure
typedef struct {
    char guestName[100];
    int roomNumber;
    char checkInDate[11];
    char checkOutDate[11];
    int Reserved;
} Reservation;

//global array
Reservation reservations[MAX_RESERVATIONS];
int reservationCount = 0;

// ALL functions
void displayMenu() ;
void addReservation();
void displayReservations();
void searchReservation();
void updateReservation();
int findByRoom();
int findByName();
void saveFile();
void loadFile();

int main() {

    int choice;
    while(1){
        displayMenu();
        printf("-----\n ");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice) {
            case 1:
                addReservation();

```

```
        break;
    case 2:
        displayReservations();
        break;
    case 3:
        searchReservation();
        break;
    case 4:
        updateReservation();
        break;
    case 5:
        saveFile();
        printf("-----Reservations saved to file.-----\n");
        break;

    case 6:
        loadFile() ;
        printf("-----file loaded-----\n");
        break ;
    case 7:
        printf("-----Exiting the program.-----\n");
        break;
    default:
        printf("Invalid choice. Please try again.\n");
}

}
```

```
// function call
```

```
void displayMenu() {
    printf("\n\n                -----");
    printf("\n\n                Hotel Room Reservation System\n");
    printf("\n\n                -----");
    printf("\n");
}
```

```

    printf("1. Add a new reservation\n");
    printf("2. Display all reservations\n");
    printf("3. Search reservation\n");
    printf("4. Update reservation\n");
    printf("5. Save reservations to file\n");
    printf("6. Load File\n");
    printf("7. Exit\n");
}

void addReservation() {
    if (reservationCount >= MAX_RESERVATIONS) {
        printf("Reservation list is full.\n");
        return;
    }

    Reservation newReservation;

    printf("Enter guest name: ");
    scanf("%s", newReservation.guestName);

    printf("Enter room number: ");
    scanf("%d", &newReservation.roomNumber);

    printf("Enter check-in date (YYYY-MM-DD): ");
    scanf("%s", newReservation.checkInDate);

    printf("Enter check-out date (YYYY-MM-DD): ");
    scanf("%s", newReservation.checkOutDate);

    newReservation.Reserved = 1;
    reservations[reservationCount] = newReservation;
    reservationCount++;

    printf("-----Reservation successful!-----\n");
}

```

```

void displayReservations() {
    if (reservationCount == 0) {
        printf("No reservations found.\n");
        return;
    }
    printf("\n-----");
    printf("\nAll Reservations\n");
    printf("-----\n");
    for (int i = 0; i < reservationCount; i++) {
        printf("Guest Name: %s\n", reservations[i].guestName);
        printf("Room Number: %d\n", reservations[i].roomNumber);
        printf("Check-in Date: %s\n", reservations[i].checkInDate);
        printf("Check-out Date: %s\n", reservations[i].checkOutDate);
        printf("Reservation Status: %s\n\n", reservations[i].Reserved ? "***
Reserved ***" : "*** Not Reserved ***");
    }
}

int findByRoom(int roomNumber) {
    for (int i = 0; i < reservationCount; i++) {
        if (reservations[i].roomNumber == roomNumber) {
            return i;
        }
    }
    return -1;
}

int findByName(char *guestName) {
    for (int i = 0; i < reservationCount; i++) {
        if (strcmp(reservations[i].guestName, guestName) == 0) {
            return i;
        }
    }
    return -1;
}

void searchReservation() {
    int choice;
    printf("Search by:\n");
    printf("1. Room Number\n");
    printf("2. Guest Name\n");
}

```

```

printf("Enter your choice: ");
scanf("%d", &choice);

if (choice == 1) {
    int roomNumber;
    printf("Enter room number: ");
    scanf("%d", &roomNumber);

    int index = findByRoom(roomNumber);
    if (index != -1) {
        printf("\n-----");
        printf("\nReservation found\n");
        printf("-----\n");
        printf("Guest Name: %s\n", reservations[index].guestName);
        printf("Room Number: %d\n", reservations[index].roomNumber);
        printf("Check-in Date: %s\n", reservations[index].checkInDate);
        printf("Check-out Date: %s\n", reservations[index].checkOutDate);
        printf("Reservation Status: %s\n", reservations[index].Reserved ?
"***Reserved***" : "***Not Reserved***");
    } else {
        printf("Reservation not found \n");
    }

} else if (choice == 2) {
    char guestName[1000];
    printf("Enter guest name: ");
    scanf("%s", guestName);

    int index = findByName(guestName);
    if (index != -1) {
        printf("\n-----");
        printf("\nReservation found\n");
        printf("-----\n");
        printf("Guest Name: %s\n", reservations[index].guestName);
        printf("Room Number: %d\n", reservations[index].roomNumber);
        printf("Check-in Date: %s\n", reservations[index].checkInDate);
        printf("Check-out Date: %s\n", reservations[index].checkOutDate);
        printf("Reservation Status: %s\n", reservations[index].Reserved ?
"***Reserved***" : "***Not Reserved***");
    } else {
        printf("Reservation not found \n");
    }
}

```

```

}

}

void updateReservation() {
    int roomNumber;
    printf("Enter the room number to update: ");
    scanf("%d", &roomNumber);

    int index = findByRoom(roomNumber);
    if (index == -1) {
        printf("-----Reservation not found----- .\n");
        return;
    }

    printf("Updating reservation for room number %d...\n", roomNumber);

    // Update guest name
    printf("Enter new guest name (current: %s): ",
reservations[index].guestName);
    scanf("%s" , reservations[index].guestName);

    printf("Enter new check-in date (YYYY-MM-DD) (current: %s): ",
reservations[index].checkInDate);
    scanf("%s" ,reservations[index].checkInDate);

    printf("Enter new check-out date (YYYY-MM-DD) (current: %s): ",
reservations[index].checkOutDate);
    scanf("%s" ,reservations[index].checkOutDate);

    printf("-----Reservation updated successfully!-----
\n");

}

void saveFile() {
    FILE *file ;
    file = fopen("hotel.txt", "w");
    if (file == NULL) {

```

```

        printf("Error opening file to save data.\n");
        return;
    }

    fprintf(file, "%d\n", reservationCount);

    for (int i = 0; i < reservationCount; i++) {
        fprintf(file, "%s,%d,%s,%s,%d\n",
            reservations[i].guestName,
            reservations[i].roomNumber,
            reservations[i].checkInDate,
            reservations[i].checkOutDate,
            reservations[i].Reserved);
    }

    fclose(file);
}

void loadFile() {
    FILE *file = fopen("hotel.txt", "r");
    if (file == NULL) {
        printf("Error opening file to load data.\n");
        return;
    }

    fscanf(file, "%d\n", &reservationCount);

    for (int i = 0; i < reservationCount; i++) {
        fscanf(file, "%99[^,],%d,%10[^,],%10[^,],%d\n",
            reservations[i].guestName,
            &reservations[i].roomNumber,
            reservations[i].checkInDate,
            reservations[i].checkOutDate,
            &reservations[i].Reserved);
    }

    fclose(file);
}

```